Maximilian Köhlenbeck

Automation Engineering student (work-study) · KUKA & Hochschule Bremen

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Profile

I study Automation Engineering in a work-study program and build systems with AI and robotics. I'm most interested in reinforcement learning, physical AI and systems that interact with the real world. Right now I'm teaching myself how to train models. I use coding agents like Claude Code and Codex for implementation and focus on system design, debugging and validation.

Education

Automation Engineering, work-study program — 10/2024 to 12/2027 (expected)
Hochschule Bremen (academic phases) and KUKA (practical phases)

Abitur (German university entrance qualification) — 2024
Gymnasium Vegesack, Bremen

Hackathons

Mobile Robotics Hackathon — 10/2026
We sorted waste with a robot arm, using 6D pose estimation and machine learning.
1st place

Hack-Nation 7 · Where Helpy started — 10/2026

Smart Region Hackathon Offenburg · Where Baulify started — 05/2026

Projects

Nozzle offset calibration for FDM 3D printing (KUKA) — 2026
My longest project so far. Errors of a few tenths of a millimeter already showed in the parts, and the offset shifted with temperature and after nozzle changes. I found a reliable, repeatable way to measure it, separated real offset errors from causes like bed leveling or material behavior, and turned it into a process the team can rerun without me. Getting consistent results took many rounds of test prints and measurements.
FDM 3D printing · Calibration · Measurement

Helpy: capturing an expert's judgment while they work (team) — 10/2026
Started at Hack-Nation 7. An assistant that watches an expert work, listens and asks about decisions at natural pauses. The result is a Work Map of steps, rules and reasons that a new colleague practices with. Screenshots are OCR'd and redacted in the browser (378 ms on a simple screen).
TypeScript · React · TanStack Start · Convex · Claude · ElevenLabs · Tesseract.js

Baulify: coordinating roadworks (team) — since 05/2026
Started at the Smart Region Hackathon Offenburg and developed further afterwards: conflict detection on a map, AI intake from PDFs and free text, approvals via a link without an account.
Next.js · TypeScript · PostgreSQL · Drizzle · MapLibre · LLM via OpenRouter

Reinforcement learning: one policy, four poses of a double pendulum — ● in progress
A single TQC policy I trained swings a double pendulum on a cart into each of its four poses and switches between them on demand. 100 % success across all 4×4 transitions after 1.7M steps, trained on a laptop CPU. It runs live in the browser on my website.
Python · Gymnasium · MuJoCo · Stable-Baselines3 · sb3-contrib (TQC) · TypeScript

Developer Island: a status capsule for the Windows desktop — 10/2026
A native app at the top of the screen for Claude Code and Codex usage, music, focus timer, meetings and system health. Reads local logs incrementally, no polling, no cloud. Idles at 55 MB and 0.03 % of one CPU core, 382 unit tests, open source.
C# · .NET 10 · WinUI 3 · SQLite · xUnit · github.com/maximilian467/Developer-Island

Skills

Reinforcement learning & simulation	Gymnasium, Stable-Baselines3 (PPO, SAC, TQC), MuJoCo
Local AI & speech	faster-whisper, Piper TTS, Ollama, ONNX / DirectML, LLMs via OpenRouter
Software & web	Python, TypeScript, JavaScript, Next.js, Astro, PostgreSQL, Drizzle, MapLibre
Automation & tooling	n8n, Claude Code, Codex, Git
Languages	German (native), English (C1)

As of October 2026